FRDM-IMXRT1186

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Revision History

Rev. Code	Date	By	Description
A	2025-03-14	Shawn Shi	Initial Release
A1	2025-05-26	Shawn Shi	1.Update DC_5V_IN and Ethernet RJ45 VCC pin. 2.Replace HyperFlash with S26HL512TFPBHM000.
A2	2025-09-05	Shawn Shi	1.Replace RGMII PHY (U21, U29) with YT8531SH-CA. 2.Populate R243 and R448. 3.Add test point for POR_CTL, CCM.CLKO1 and CCM.CLKO2. 4.Replace CAN with TJA1057BT. 5.Replace R119 and R502 with 10ohm, replace R487 with 33ohm based on prescan test. 6.Add an inverter and pull up resistor for ECAT_RESET_OUT 7.Add test point for automated test platform.
A3	2025-10-21	Shawn Shi	DNP R30 and R31, added Arduino interface I2C note on page18. No layout change.
A4	2025-12-02	Shawn Shi	No layout change 1. Correct typo in the block diagram. 2. Change J4 from 1-2 to 5-6. 3. Improve statement boot switch default set up.

Jumper Setting

REF DES	JUMPER(DEFAULT)	PAGE NAME
J5	1-2	03 Main Power
J4	5-6	03 Main Power
J10	1-2	04 Power Domain
J15,J16,J19,J20,J21,J38	1-2	05 MIMXRT1180 Part1
J11,J37	Open	05 MIMXRT1180 Part1
J27,J28,J29,J30,J31,J32,J33, J35,J41,J43,J44,J45,J46,J47, J48	1-2	06 MIMXRT1180 Part2
J42,J49	2-3	06 MIMXRT1180 Part2
J12,J13,J17,J18	2-3	07 MIMXRT1180 Part3
J8,J24,J26	1-2	16 Boot & Reset
J7,J9,J14,J25	Open	16 Boot & Reset
J34,J59	1-2	17 MCU-LINK
J36,J40,J50,J58	Open	17 MCU-LINK
J55	Open	18 Arduino/JTAG/SRAMC

Switch Setting

REF DES	SWITCH(DEFAULT)	PAGE NAME
J60	off.off, on	16 Boot & Reset

1. Unless Otherwise Specified:

   All resistors are in ohms, 1/16 Watt,0402
   All capacitors are in uF,0402
   All voltages are DC
   All polarized capacitors are aluminum electrolytic

2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

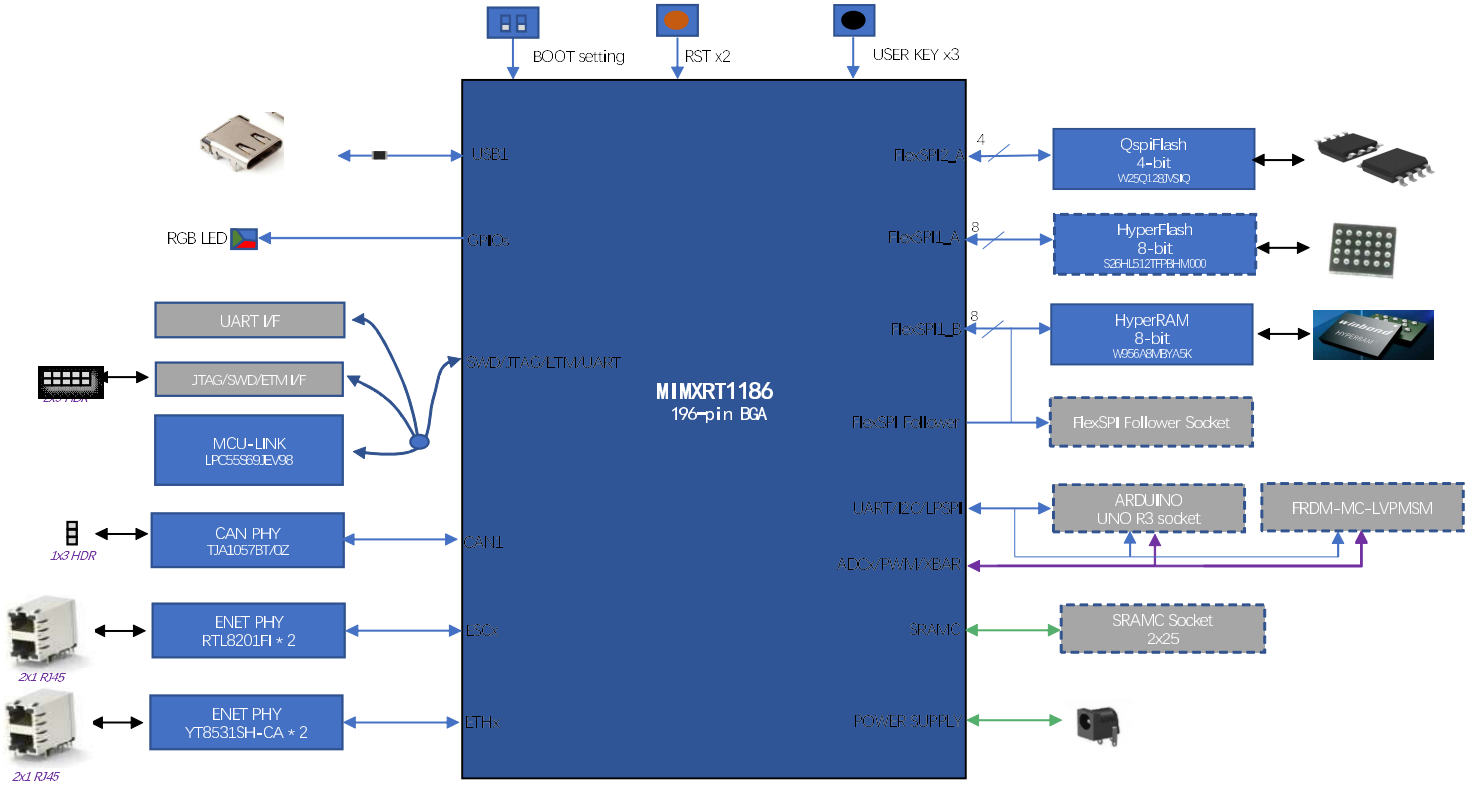
3. Device type number is for reference only. The number varies with the manufacturer.

4. Special signal usage:
   _B Denotes - Active-Low Signal
   <> or [] Denotes - Vectored Signals

5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

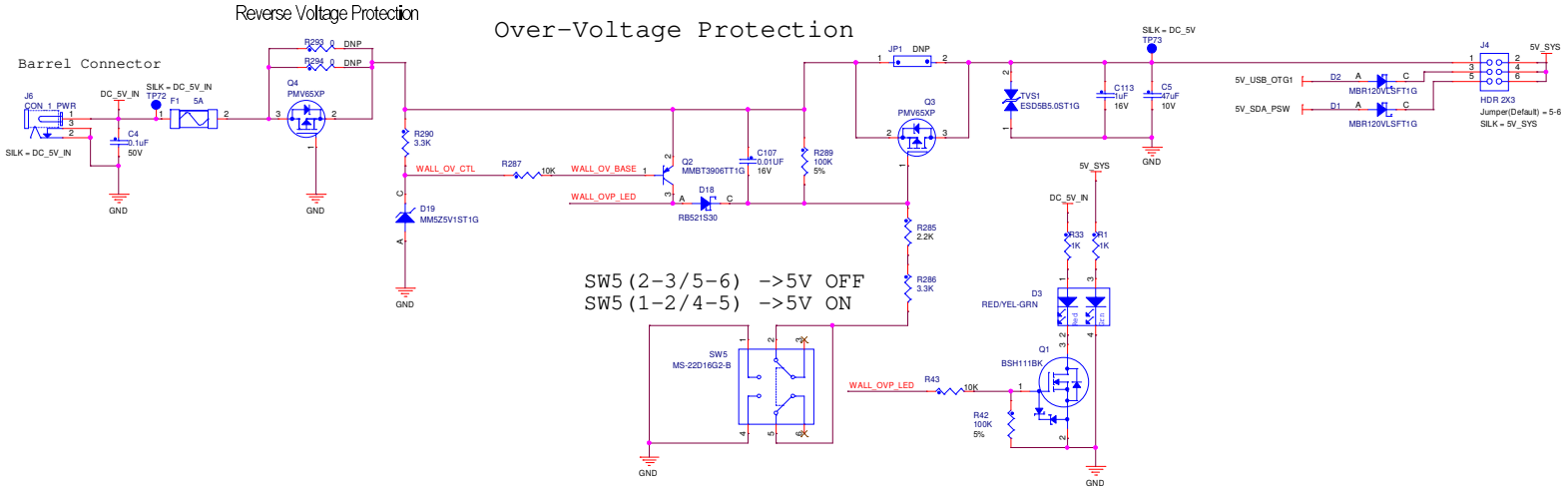
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Drawing Title: FRDM-IMXRT1186
Page Title: Cover
Size C | Document Number SCH-95302, PDF:SPF-95302 | Rev A4
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FRDM-IMXRT1186

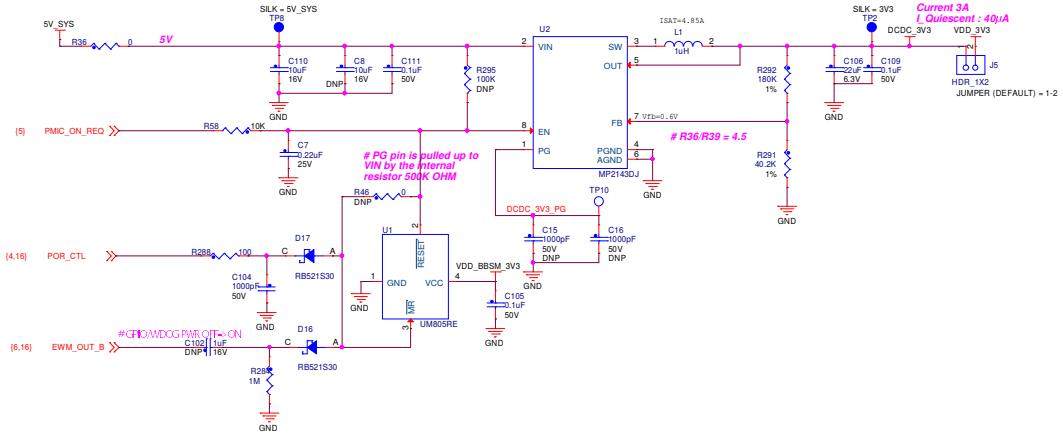


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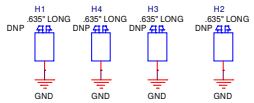
Main Power



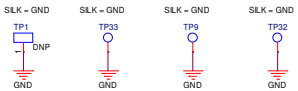
5V To 3.3V



Board Mounting Holes



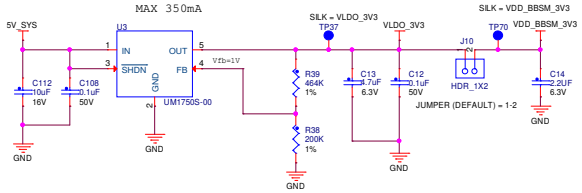
Ground TPs



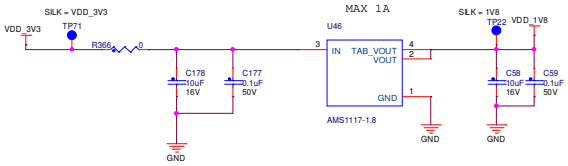
Layout Note: Place Ground TPs to assist signal measurement.

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Page Title:	Main Power
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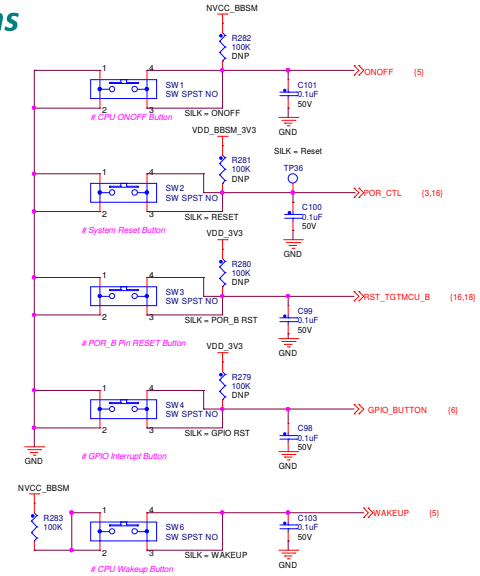
3V3 LDO for BBSM



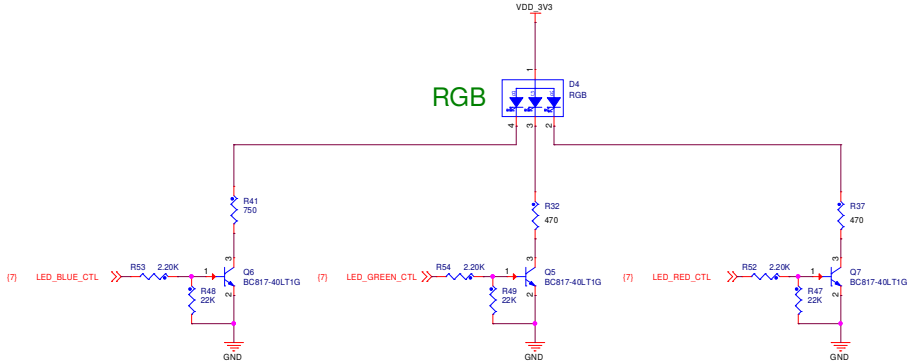
1V8 LDO



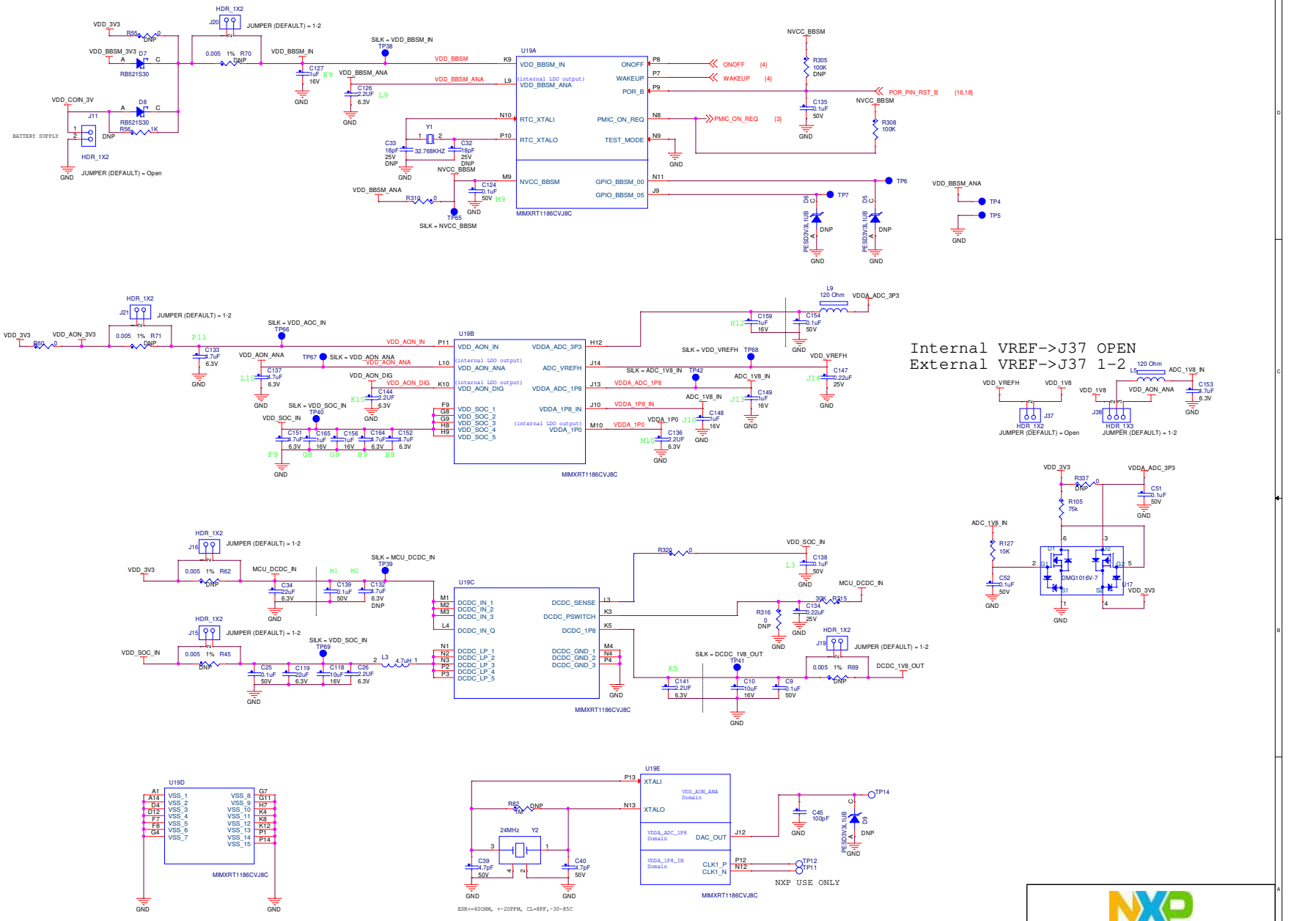
Buttons



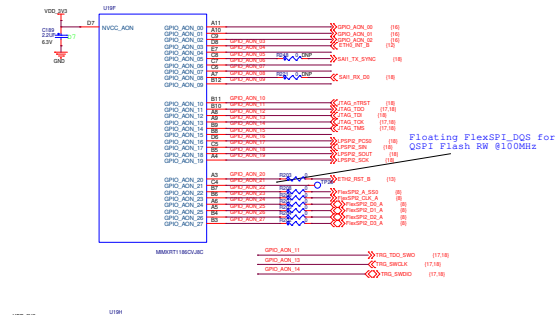
User LED

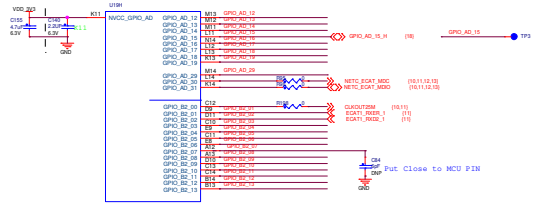


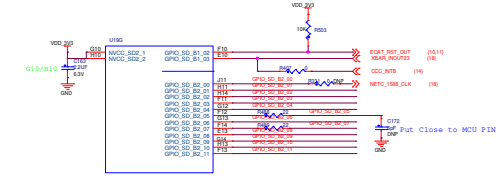
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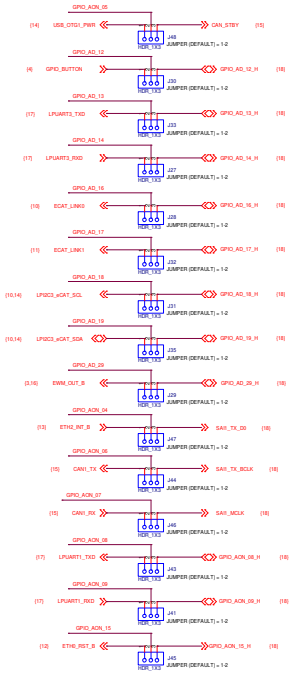
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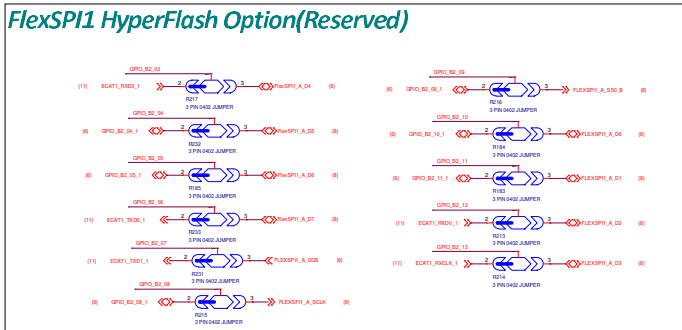
Function PinMux



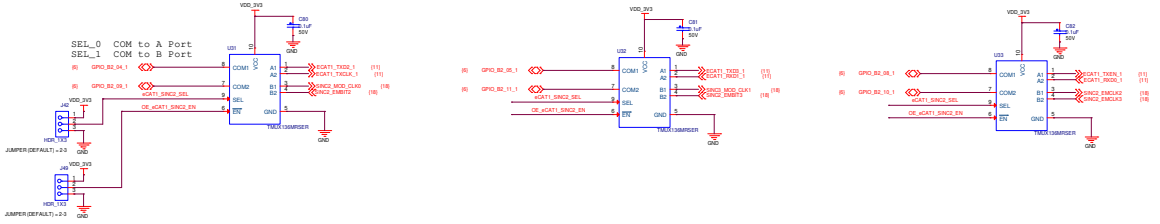
FlexSPI Follower Option(Reserved)



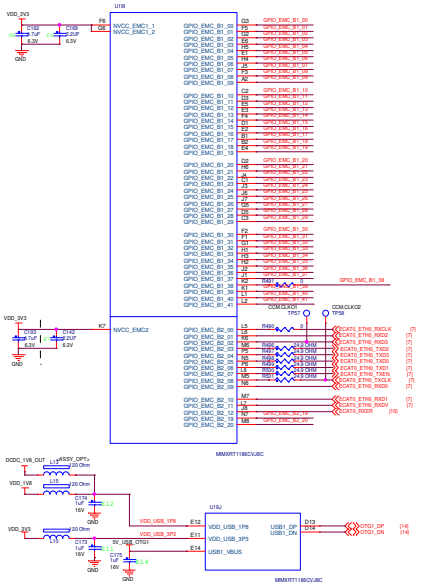
FlexSPI1 HyperFlash Option(Reserved)



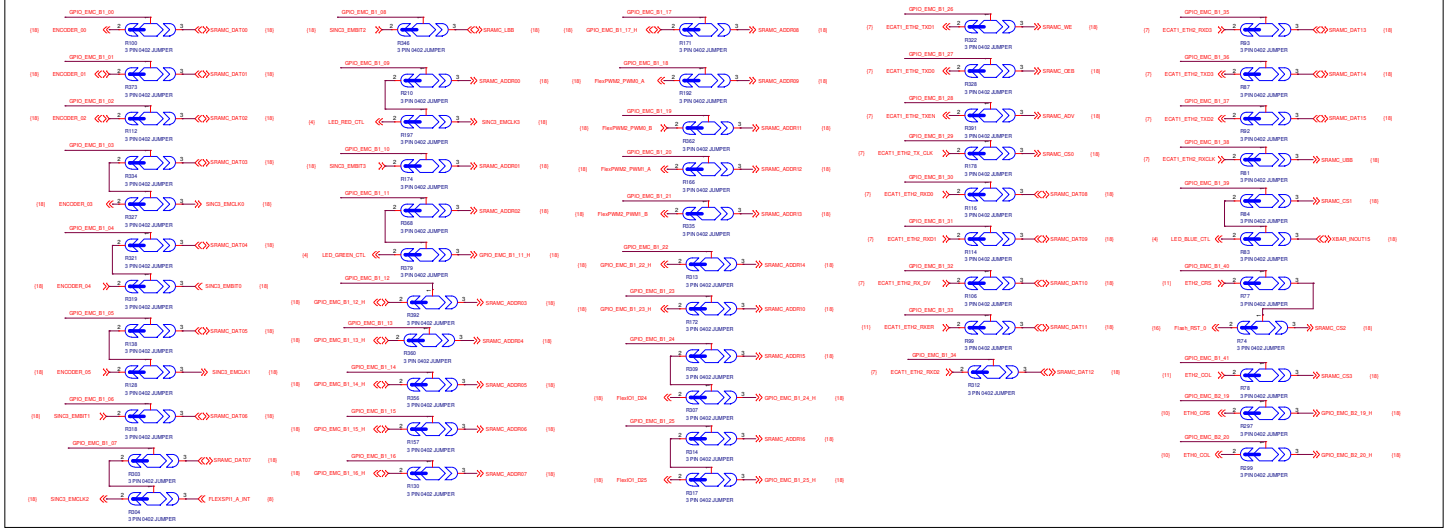
J42(2-3)/J49(2-3): eCAT Function Pin (Default)
J42(1-2)/J49(2-3): SINC Function Pin



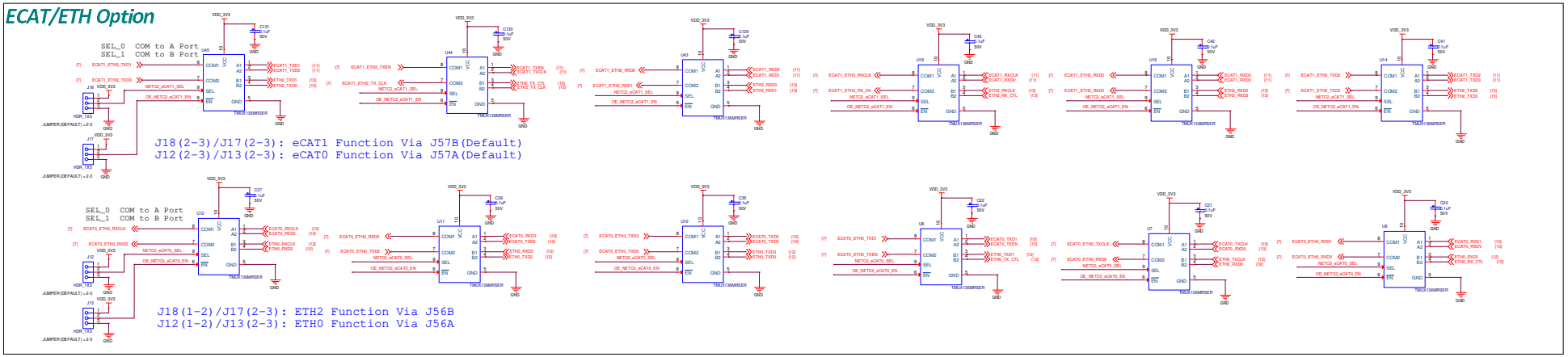
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Size: B	Document Number: SCH-69302_PDF-SPF-69302
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SRAMC Option(Reserved)

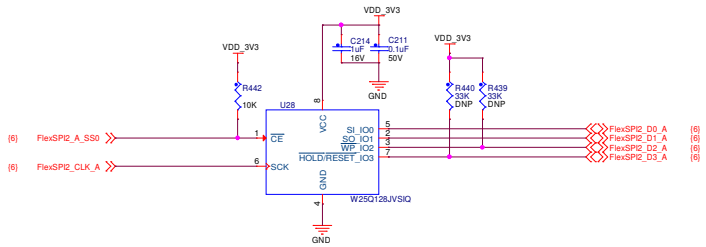


ECAT/ETH Option



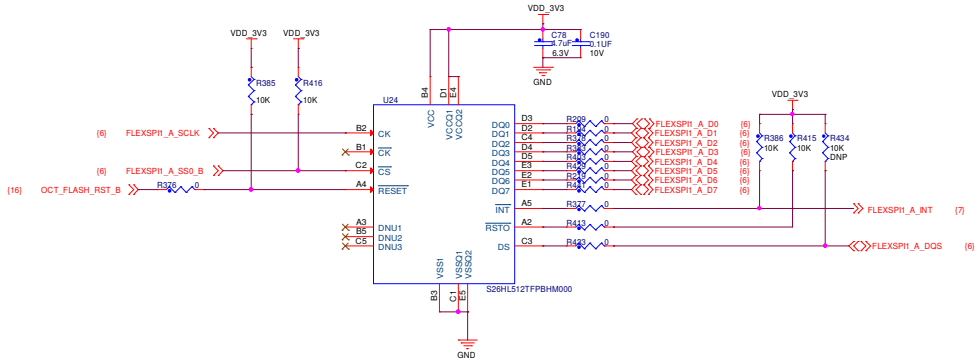
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QSPI Flash



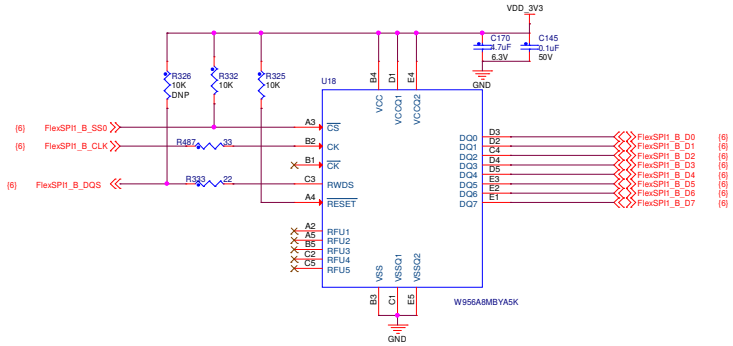
HyperFlash(Reserved)

To Enable HyperFlash:
R183~R185, R213~R217, R231~R233 Change to 1-3



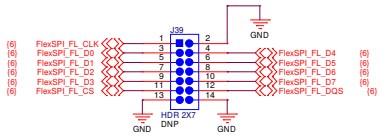
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Size: C	Document Number: SCH-95302, PDF:SPF-95302	Rev: A4
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HyperRAM



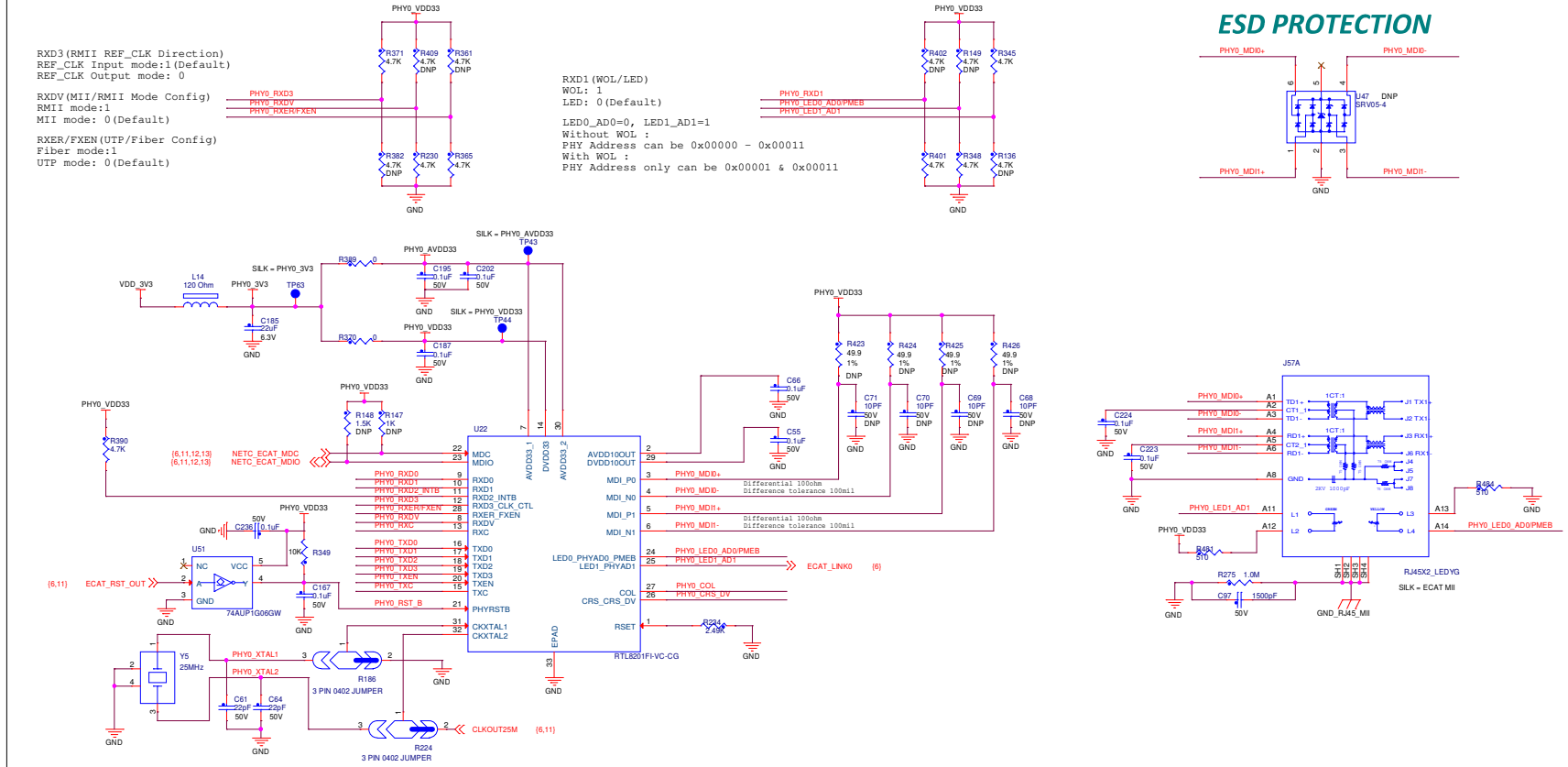
FlexSPI Follower(Reserved)

To Enable FlexSPI Follower:
R113, R117, R131~R133, R135, R160~R163, R182 Change to 1-3



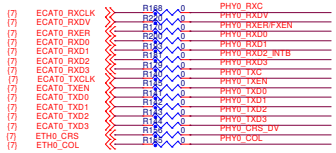
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MII ECAT0 {J12(2-3)/J13(2-3): eCAT0 Function Via J57A(Default)}

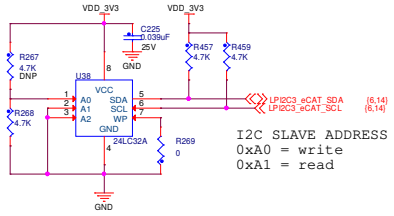


ESD PROTECTION

MII Interface

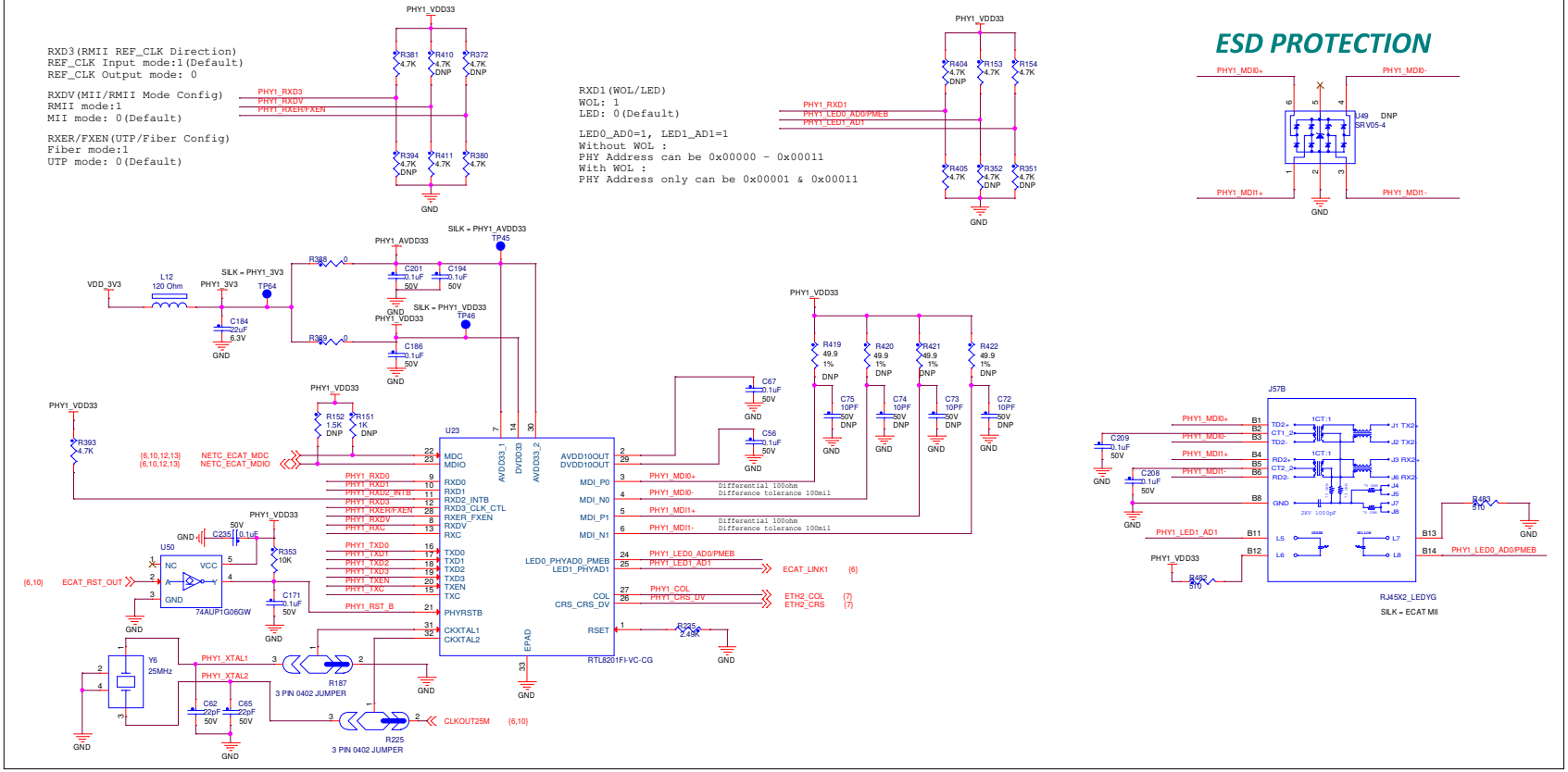


EEPROM



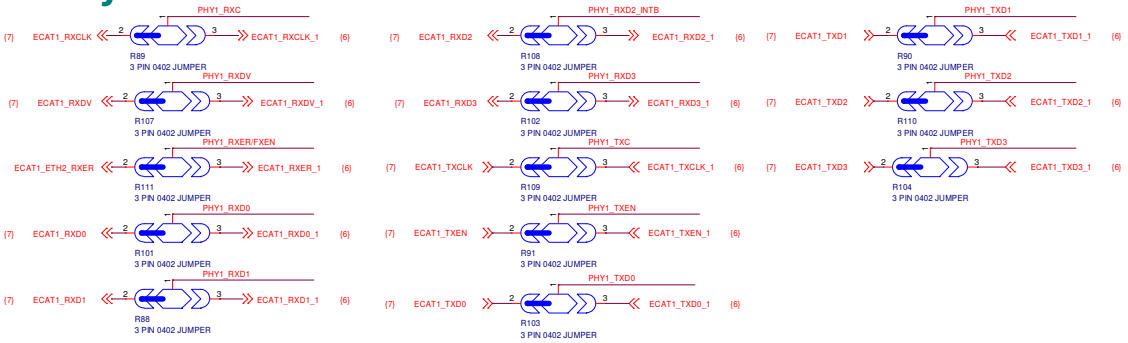
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Size: C	Document Number: SCH-95302, PDF:SPF-95302		Rev: A4
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MII ECAT1 {J18(2-3)/J17(2-3): eCAT1 Function Via J28B(Default)}



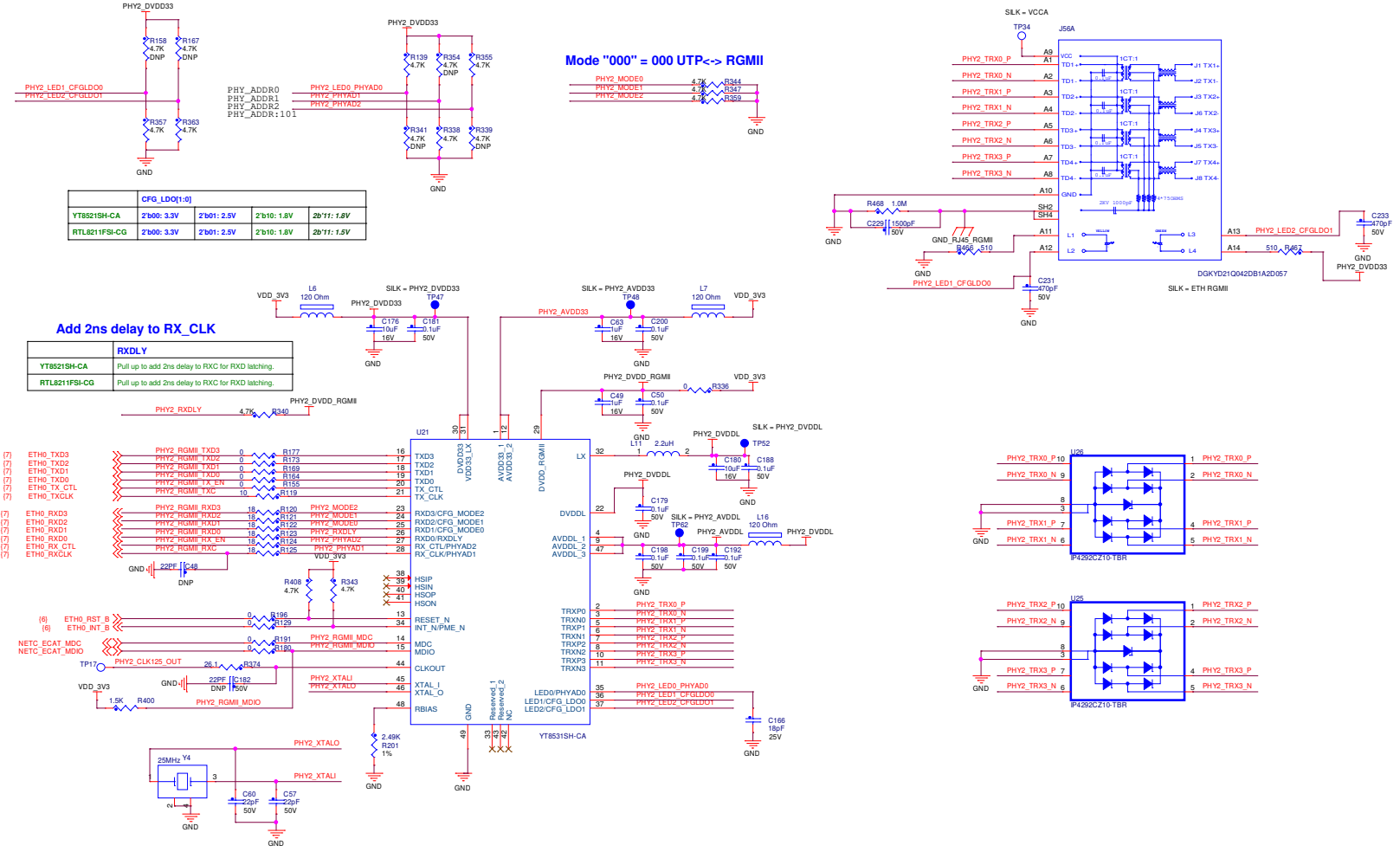
ESD PROTECTION

MII Interface



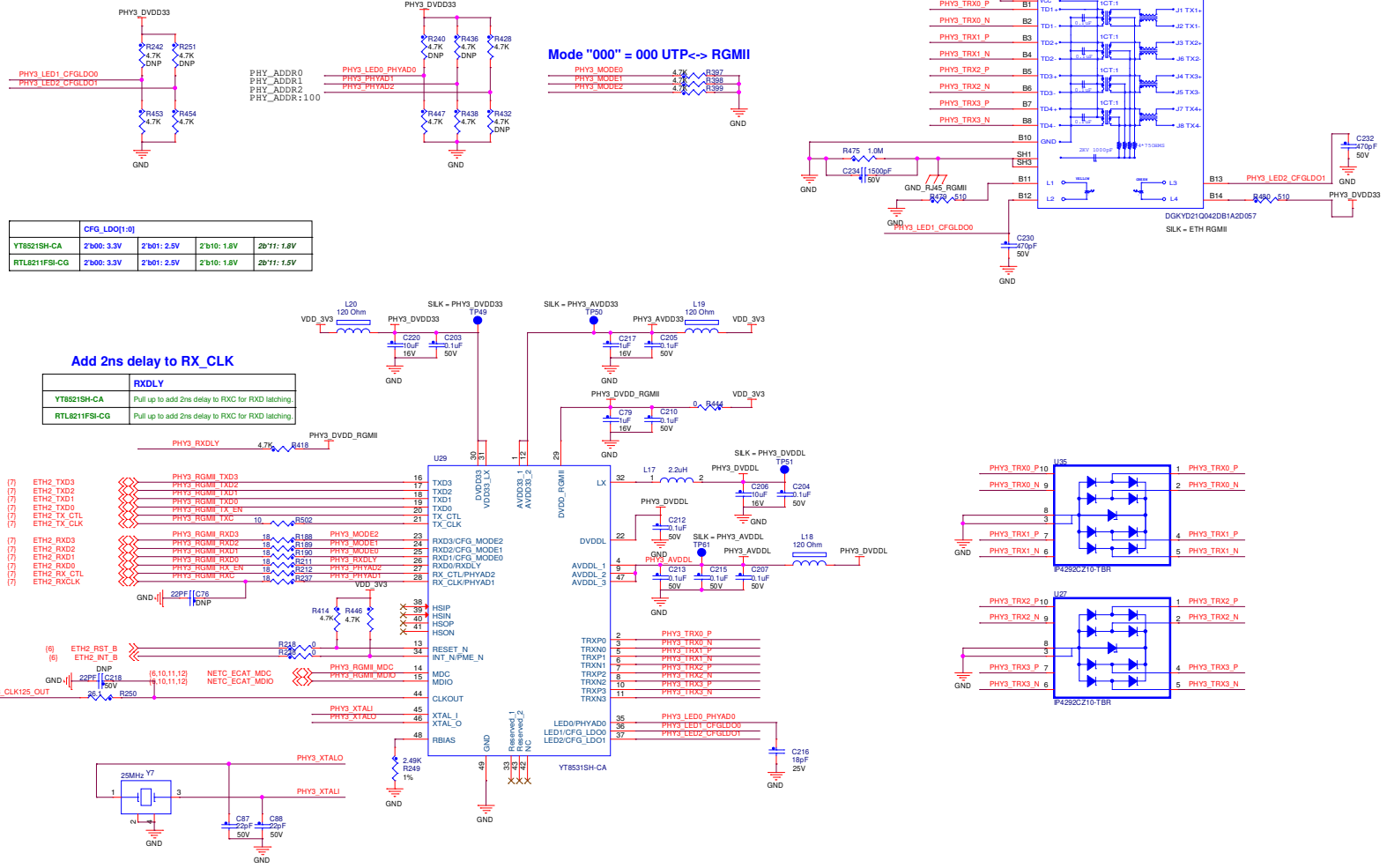
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Size: C	Document Number: SCH-95302, PDF:SPF-95302 Rev: A4
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RGMII ETH0 {J12(1-2)/J13(2-3): ETH0 Function Via J56A}



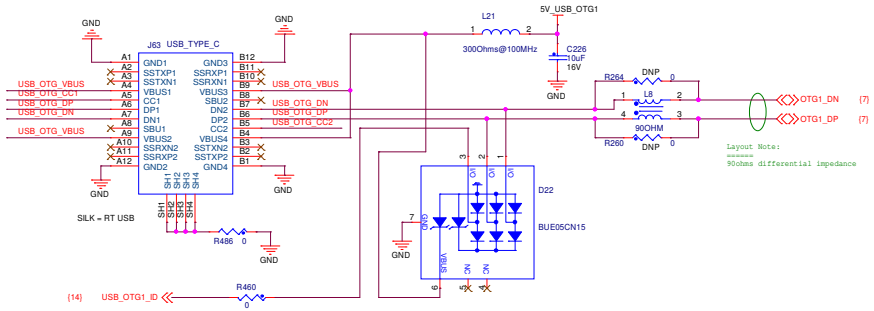
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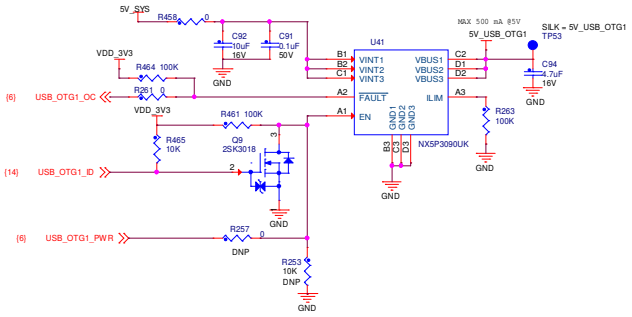
RGMII ETH2 {J18(1-2)/J17(2-3): ETH2 Function Via J56B}



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USB OTG1

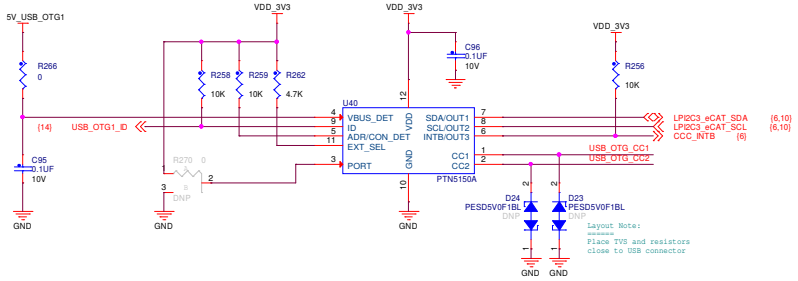




USB ID is a processor Analog Sense Input

Host --> ID = GND
Device --> ID = Floating

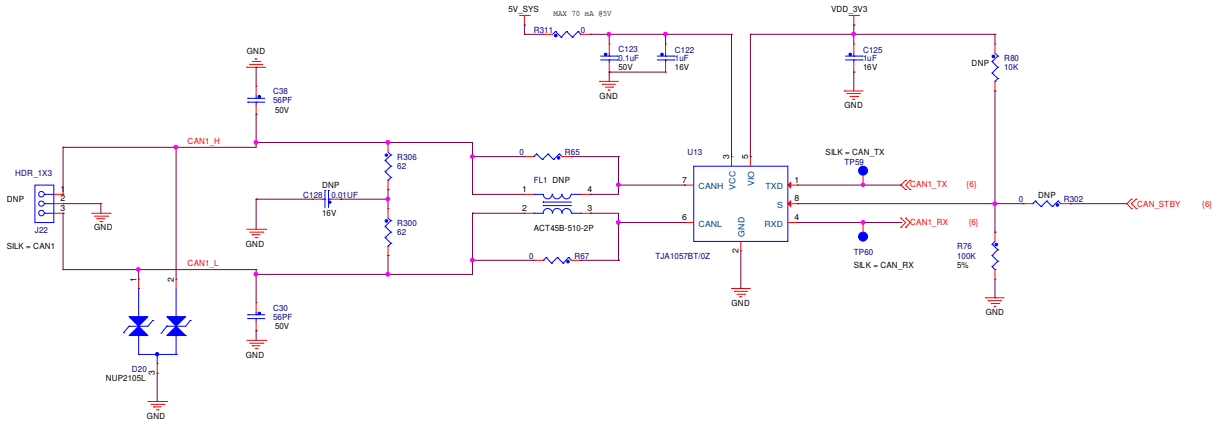
CC Logic



EXT_SEL	PORT	ADR/CON_DET
0---CC2 1---CC1	PORT=1: DFP mode PORT=0: UFP mode PORT=Float: DRP mode	CON_DET=1, Connection Detected, I2C Address: 0x7A CON_DET=0, No Connection, I2C Address: 0x3A
Default 1	PORT=Float: DRP mode	Float, no I2C function

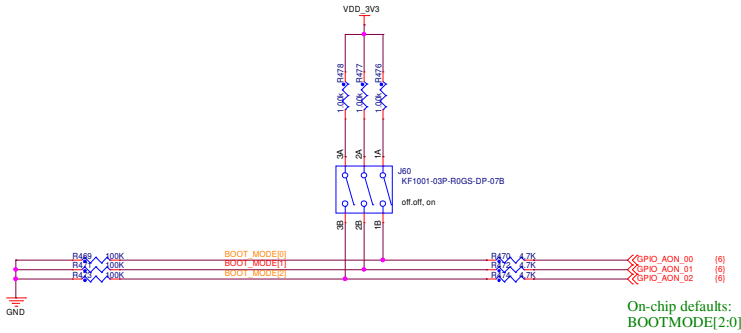
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CAN BUS



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External Boot Switch

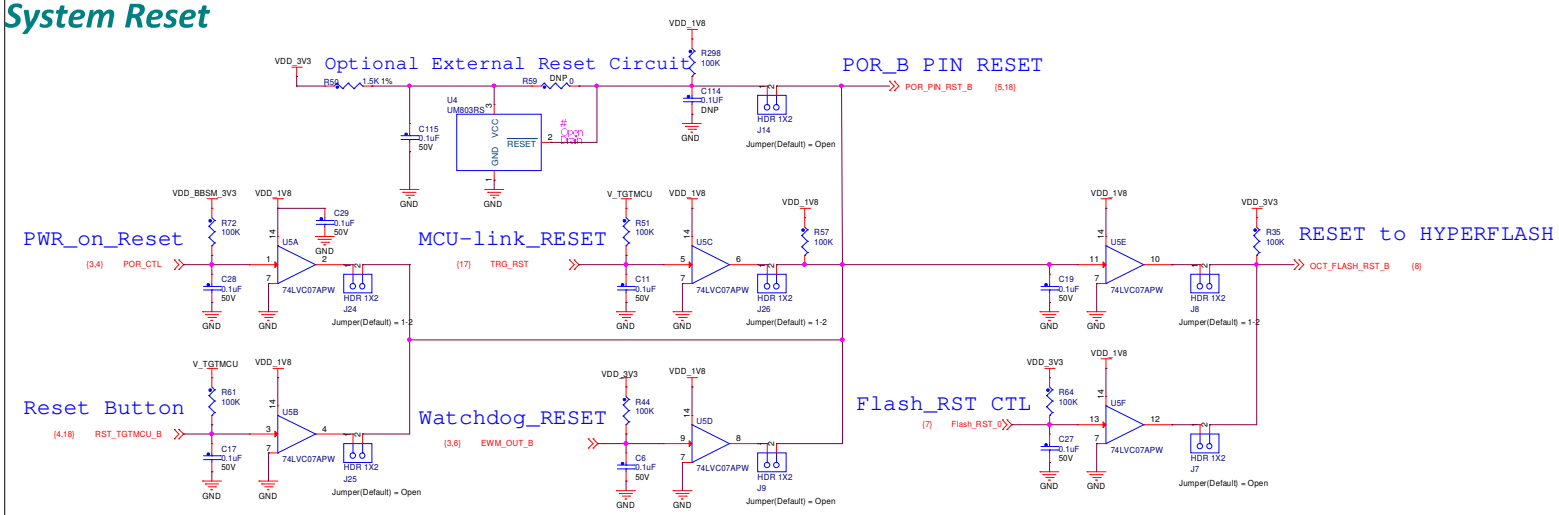


On-chip defaults:
BOOTMODE[2:0] = 25-50 kohm PD

Boot Mode pin settings

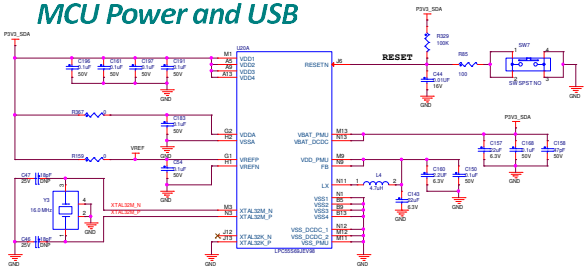
BOOT_MODE[2:0]	Boot Type
000(J60-x000)	Boot from Internal Fuses
001(J60-x001)	Serial Downloader (USB1 or LPUART1)
010(J60-x010)	USDHC2 8-bit 1.8V eMMC 5.1
011(J60-x011)	uSDHC1 4-bit SD 3.0
100(J60-x100)	FlexSPI Quad SPI Serial NOR (or Octal SPI NOR in Quad SPI NOR mode)
101(J60-x101)	FlexSPI Quad SPI Serial NAND 2k page (or Octal SPI NAND in Quad SPI NOR mode)
110(J60-x110)	FlexSPI Quad SPI Serial NAND 4k page (or Octal SPI NAND in Quad SPI NOR mode)
111(J60-x111)	Test Mode

System Reset

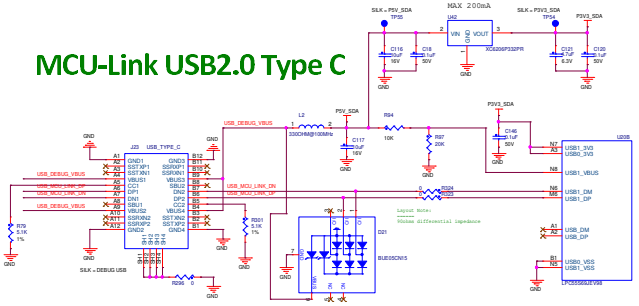


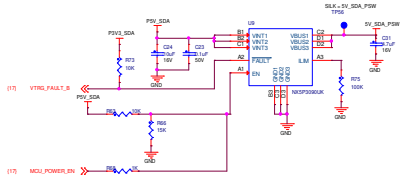
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Page Title:	Boot & Reset	
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MCU Power and USB



MCU-Link USB2.0 Type C





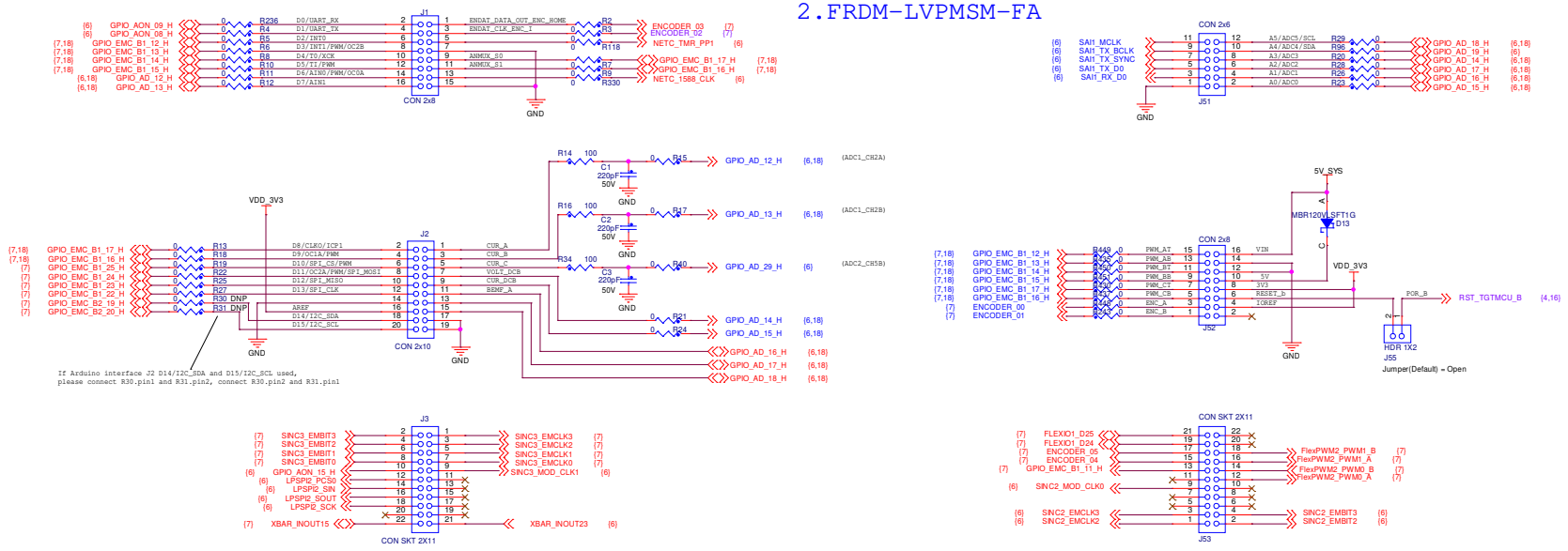


Arduino & MC Interface

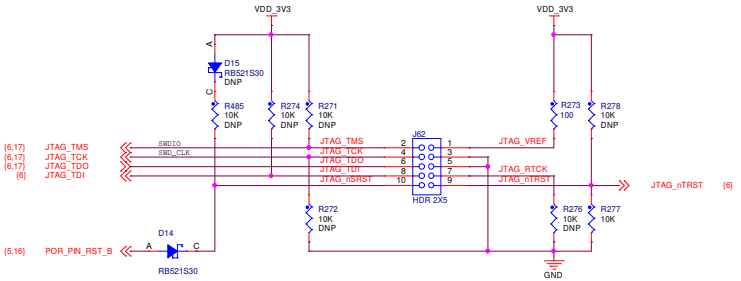
Motor Card Supported:

1.FRDM-MC-LVPMSM

2.FRDM-LVPMSM-FA

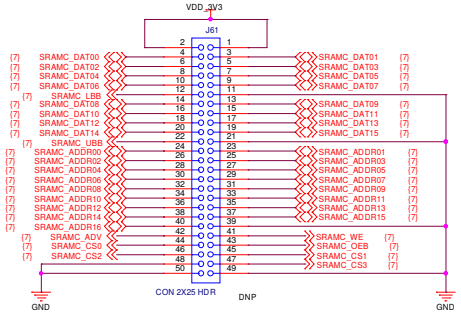


JTAG(SWD)



SRAMC Interface (Reserved)

To Enable SRAMC:
Related Jumper Wiring Change to 1-3



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